COSC 428.100, COSC 528.100 **Bowie State Univ.**
Design and Analysis of Algorithms **Arts & Science**
3 Credits
Fall 2026

Syllabus

ADA Statement

Bowie State University is committed to creating inclusive and accessible learning environments consistent with federal and state law. If you have registered with Disability Support Services (DSS) and an accommodation notification was sent on your behalf, please schedule a meeting with me to discuss how your accommodations will be implemented in this course.

Students who have a permanent disability or temporary health condition that requires accommodations should contact Disability Support Services immediately. You may email: dss@bowiestate.edu , call 301-860-4085 or stop by their office: Thurgood Marshall Library, lower-level, RM# 078.

Our website provides guidance on how to register with Disability Support Services: https://www.bowiestate.edu/academics/support-services/disability-support-services/request-services/

Important Dates

Last Day to Add/Drop a Class	09/04/26
Mid-Semester Period	10/12-16/26
Last Day to Change to AU	10/30/26

Instructor:

Jim Gil de Lamadrid
CSB 222
Phone: 860-3968
jgildelamadrid@bowiestate.edu

Office Hours:

MW 2:00-3:00p
TR 2:30-4:30p

Prerequisites:

undergraduate: COSC 214

Text:

McConnell, J. J., *Analysis of Algorithms: An Active Learning Approach, **2nd ed.***, Jones an Bartlett Publishers, Inc., Sudbury, Massachusetts, 2001.

Instruction Modes:

The course will be taught as a series of lectures. Students will also complete written exercises, and a programming project with a presentation.

Description:

This course will include the design and analysis of algorithms. Algorithms that manipulate data structures such as lists, stacks, queues, trees, and graphs are discussed in terms of efficiency and existence. Algorithms for data compression, memory management, and advanced hashing techniques will be studied. Topics also include balancing algorithms, divide-and-conquer algorithms, dynamic programming, the greedy

approach, backtracking, branch-and-bound, advanced search/sort techniques, and P=NP problem, parallel algorithms, and newly promoted algorithms will be studied and researched.

Objectives:

Upon completion of this course students will

1. be able to apply their knowledge an of issues of efficiency, and the limitations of computation in problem analysis,†
2. have been provided with the tools necessary for analyzing a programs efficiency,††
3. possess familiarity with strategies for solving some of the more fundamental computation problems.
4. Be able to communicate about efficiency concerns.†††
5. participate in producing a group product.††††

Requirements:

Group project	10.00%
Homeworks (6)	40.00%
Tests (2, noncumulative)	50.00%

Grading:

Students are required to do all work individually. Collaboration resulting in plagiarism is considered cheating, and will be subject to discipline. Disciplinary action may consist of a failing grade on the effected assignment, a failing grade in the course, or more severe discipline impose by the University scholastic standards committee.
Homework and tests will be graded on an absolute scale, although the final score may be curved at the instructor's discretion. The curve will never deflate your grade with respect to the absolute scale. The scale used is:

A	100%-90%
B	89%-80%
C	79%-70%
D	69%-60%
F	59%-0%

Homework is due on time. Late homework will be accepted, but penalized. Late homework is worth 80% of its original value. Late homework may be turned in until and through the class a week after the class in which it is due. Any homework turned in later than that, or after the scheduled final exam date, will not be accepted. All homework is due at the beginning of the class in which it is due. In the event of unforeseen problems, requiring a cancellation of class, any tests scheduled, during a canceled class will be rescheduled for the next class.

Students are responsible for all content delivered in lecture. This includes any announcement, whether or not it they are posted in writing. Attendance is required. Students are allowed three absences during the semester. The final score for a student that has accumulated more than the three absences will be decreased by 10%.

Programming assignments will be scored using the following categories:

Style - Comments, blank lines, line spacing, indentation, variable names.
Syntax - Does the program compile.
Semantics - Does the program produce correct results.
Design - Is the design unduly complex

Written work will be scored on clarity, organization, thoroughness, and completeness (whether everything requested is present).

References:

Brassard, G., Bartley, P., *Fundamentals of Algorithmics*, Prentice Hall, Inc., Englewood Cliffs, New Jersey, 1996.

Cormen, T. H., Leiserson, C. E., Rivest, R. L., *Introduction to Algorithms*, MIT Press, Cambridge, Massachusetts, 1990.

Sedgewick, R., *Algorithms in C*, Addison-Wesley Publishing Co., 1990.

Outline:

Week #1		Introduction, Math Background	pp.1-19
Week #2	Hmwk. #1	Order of Growth, Recurrence Relations, Divide-and-Conquer	pp.19-42
Week #3		Searching, Selection	pp.59-76
Week #4	Hmwk. #2	Basic In-Place Sorting	pp.77-94
Week #5		Radix, Heap, and Merge Sorts	pp.94-113
Week #6	Hmwk. #3	Quicksort, Polyphase Merge Sort	pp.113-130
Week #7	Test #1		
Week #8	Project	Basic Graph Theory	pp.231-244
Week #9	Hmwk. #4	Minimum Spanning Tree, Shortest Path	pp.245-258
Week #10		Connected Components, Partitioning	pp.258-266
Week #11	Hmwk. #5	Parallel Algorithms	pp.267-288 pp.297-302
Week #12		Greedy and Dynamic Algorithms	pp.355-365 pp.385-396
Week #13	Hmwk. #6	Presentations	
Week #14		P=NP	pp.338-354
Week #15	Test #2		

ABET Material:

Corresponds to ABET Program outcomes:

† 1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.

†† 6.1. Demonstrates and applies computer science theories to solve a computing problem.

6.2. Recognizes an appropriate algorithm to solve a problem.

††† 3. Communicate effectively in a variety of professional contexts.

†††† 5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.